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10 MARCH 1959
SUPERSEDING
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23 DECEMBER 1954

MILITARY SPECIFICATION

PANEL, TELEPHONE SWITCHBOARD (PANEL BD-132( ))

This specification has been approved by the Department of Defense and is mandatory for use by the Departments of the Army, the Navy, and the Air Force.

1. SCOPE

1.1 This specification covers a portable power control and alarm panel designated Panel BD-132( ).

2. APPLICABLE DOCUMENTS

2.1 The following specifications, standards, and drawings of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

FEDERAL

QQ-S-571 — Solder; Soft (Tin, Tin-Lead, and Lead-Silver).

UU-T-111 — Tape, Paper, Gummed (Sealing and Securing).

LLL-F-291 — Fiberboard; Corrugated, Single Face (Flexible).

PPP-B-621 — Boxes, Wood, Nailed and Lock-Corners.

PPP-B-636 — Boxes, Fiber.

PPP-B-60 — Tape; Pressure Sensitive Adhesive, Waterproof — for Packaging and Sealing.

MILITARY

MIL-P-116 — Preservation, Methods of.

MIL-B-117 — Bags, Interior Packaging.

MIL-B-121 — Barrier Material, Greaseproofed, Flexible (Waterproofed).

MIL-M-11473 — Magnetic-Particle Inspection; Soundness Requirements for Weldments.

MIL-S-11748 — Suppression Requirements for Electrical and Electronic Equipment (Except Internal Combustion Engine-Driven Equipment).

FSC 5805

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STANDARDS

MILITARY

MIL-STD-105 — Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-129 — Marking for Shipment and Storage.

MIL-STD-130 — Identification Marking of U.S. Military Property.

MIL-STD-169 — Extreme Temperature Cycle.

MIL-STD-170 — Moisture Resistance Test Cycle for Ground Signal Equipment.

MIL-STD-252 — Wired Equipment, Classification of Visual and Mechanical Defects for.

DRAWINGS

SIGNAL CORPS

SC-DL-90699 — Panel BD-132( ) (Power).

(Copies of specifications, standards, and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer. Both the title and number or symbol should be stipulated when requesting copies.)

2.2 *Other publications.* The following documents form a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

AMERICAN TRUCKING ASSOCIATION

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Association, 1424 16th St., N.W., Washington 6, D. C.)

ASSOCIATION OF AMERICAN RAILROADS

Uniform Freight Classification.

(Application for copies should be addressed to the Association of American Railroads, 202 Chicago Union Station, Chicago 6, Ill.)

3. REQUIREMENTS

3.1 **Description.** Panel BD-132( ) is the connecting point for the 48-volt battery circuits commonly associated with telephone central office equipment. The panel is also the source of ringing power for the switchboard, or switchboards, in a central office setup. The panel is equipped with the necessary meters, switches, and other components required to supply and control power needed for proper operation of a telephone central office.

3.2 **Construction.** Panel, Telephone Switchboard (Panel BD-132( )) shall be constructed in accordance with the drawings listed in Drawing and Data List SC-DL-90699.

3.3 **Preproduction samples.** The contractor shall furnish preproduction samples for approval, as required by the invitation for bids and contract (see 6.2).

3.4 **Parts, materials, and processes; general.**

3.4.1 *Parts and materials.* Except as otherwise specified herein, parts and materials used in fabrication of the equipment shall be as specified in the drawings covered by Drawing and Data List SC-DL-90699.

3.4.2 *Processes.*

3.4.2.1 *Cleaning.* Metal parts, after fabrication, shall be cleaned in accordance with good commercial practice, or as specified in an applicable document. Cleaning processes shal have no deleterious effect on the equipment. Corrosive material shall be removed completely before parts are assembled into

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the equipment. After assembly, components shall be cleaned thoroughly and shall be free from particles of solder, flux, and other foreign material. If necessary, such cleaning shall also be performed before final assembly of components.

3.4.2.2 *Riveting.* After riveting, the joints shall be tight, the joined parts undamaged and the rivet heads properly seated and tight against their bearing surfaces.

3.4.2.3 *Soldering.*

3.4.2.3.1 *Solder.* Solder used for electrical connections shall be composition Sn60 conforming to Specification QQ–S–571.

3.4.2.3.2 *Flux and cleaning agents.* No acid or acid salts shall be used in preparation for or during soldering; however, exception is permitted for preliminary tinning of electrical connections and for tinning or soldering of mechanical joints not used to complete electrical circuits, but in no case shall acid or acid salts be used where they can come in contact with insulation material. Where acid or acid salts are used, as permitted above, they shall be completely neutralized and removed immediately after use. Flux for soldering of electrical connections shall be rosin, rosin and alcohol, or rosin and turpentine.

3.4.2.3.3 *Process.* There shall be no sharp points or rough surfaces resulting from insufficient heating. The solder shall feather out to a thin edge, indicating proper flowing and wetting action, and shall not be crystallized, overheated, or underheated. The maximum necessary amount of flux and solder shall be used for electrical connections. Any means employed to remove an unavoidable excess of flux shall not incur the risk of loose particles of flux, brush bristles, or other foreign material remaining in the equipment; flux being spread over a larger area; or damage to the equipment. Insulation material that has been subjected to heating during the soldering operation shall be undamaged and parts fastened thereto shall not have become loosened.

3.4.2.4 *Welding.* Welds shall be of ample size. Where tack or spot welds are employed, the number of welds shall be sufficient to provide adequate strength. Spot or tack welds in aluminum or magnesium shall be limited to lightly stressed, noncritical structures.

3.4.2.4.1 *Cleaning.* Surfaces to be welded shall be free from rust, scale, paint, grease, and other foreign material. Welded assemblies shall be cleaned of scale, oxidation products, and excess flux by sandblasting, wire brushing, or other suitable means. Acid used for cleaning shall be completely neutralized and removed.

3.4.2.4.2 *Process.* Preheating shall be employed where distortion is likely to result from welding. Welds shall have thorough penetration and good fusion and shall be free from scals, blisters, abnormal pock marks, and other harmful defects. Where undesirable internal stresses are likely to result from welding, the welded assembly shall be a stress-relieved. Inert-gas shielded arc welding shall be used, when practicable, for welding of aluminum or magnesium.

3.4.2.4.3 *Nonvisual defects.* Welding, when subjected to the test of 4.10, shall evidence good fusion and shall be free of cracks, voids, and slag inclusions.

3.4.2.5 *Wiring and cabling.*

3.4.2.5.1 *Slack.* Wires and cables shall be as short as practicable except that sufficient slack shall be provided to prevent undue stress on cable forms, wires, and connections; to enable parts to be removed and replaced during servicing without disconnecting other parts; and to facilitate field repair of broken or cut wires.

3.4.2.5.2 *Protection.* Wires and cables shall be so placed and protected as to avoid

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contact, under specified service conditions, with rough or irregular surfaces or sharp edges. Wires shall not be bent sharply where they enter insulation material. Where wires run through holes or slots in metal (partitions, shields, and similar items), they shall be protected by suitable grommets or bushings.

3.4.2.5.3 *Support.* Wires and cables shall be properly supported and secured, to prevent undue stress on conductors and terminals and undue change in position of the wires and cables after the equipment has been subjected to specified service conditions or has been serviced or repaired in a normal manner.

3.4.2.5.4 *Clearance.* Clearance between wires and cables, and parts, shall be sufficient to avoid deterioration of the wires or cables because of the heat dissipated by such parts when the equipment is subjected to specified service conditions. Clearance between solder connections or bare conductors, on terminal boards, relays, or other parts, shall be at least ¼ inch unless use of specified parts makes such clearance impracticable.

3.4.2.5.5 *Connections.* Before being soldered to terminal lugs or fixed terminals, wires shall be mechanically secured so that the connections are not dependent for strength on solder alone. Electrical connections shall not be made by clamping between a metallic and a nonmetallic material. Fraying of textile ends of wires shall be prevented mechanically or by application of approved varnish.

3.4.2.5.6 *Terminals, fixed.* Fixed terminals (including binding posts) shall properly support the wires connected thereto and shall not turn or become loose when the equipment is subjected to specified service conditions.

3.5 Electrical requirements.

3.5.1 *ac ringing converter.*

(a) With a 60-cycle input frequency, the output frequency shall be 18-22 cycles.

(b) With a 60-cycle, 115-volt rms input, the output into a noninductive circuit shall be as required below. The resistance of the output circuit shall include that of the measuring instrument, which shall be of the thermocouple or equivalent type to avoid errors due to waveform.

Resistance	*Minimum output*
200 ohms ± 5 percent	140 milliamperes
1,000 ohms ± 5 percent	70 milliamperes
10,000 ohms ± 5 percent	10 milliamperes

(c) With a 60-cycle, 115-volt rms input, the peak output voltages shall not exceed 250 volts.

3.5.2 *dc ringing converter.*

(a) *Input.*

(1) The dc ringing converter shall be capable of continuous operation (one hour minimum) from a storage battery having voltage limits of 44–56 volts.

(2) The nonload input shall not exceed 50 watts.

(3) The starting current shall not exceed 7.5 amperes for 0.5 second at no load output.

(b) *Output.* With an input voltage of 48 volts, dc the output shall be as follows:

(1) The output frequency shall be 18 to 22 cycles per second.

(2) The no-load output rms voltage shall not be less than 90 volts.

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(3) The no-load output peak voltage shall not exceed 250 volts.

(4) The output rms voltage shall be not less than 70 volts across terminals of 200 ohms ± 5 percent non-inductive resistance.

3.5.3 *Circuit breakers.* Circuit breakers shall hold on 100 percent rated load. Circuit breakers shall trip at 125 percent rated load, in 8 seconds for alternating current (ac) breakers and in 2 seconds for direct current (dc) breakers in accordance with 4.5.3. Circuit breakers shall trip at 300 percent rated load in 0.4 second for ac breakers and in 0.3 second for dc breakers in accordance with 4.5.3.

3.5.4 *Breakdown voltage.* Air gaps (in relays, switches, and similar items) and electrical insulation shall be capable of withstanding 500 volts without any visible deterioration or degradation in specified performance requirements. The test voltage shall be dc or peak ac of the same value and shall be applied for 10 seconds. If ac, the waveform shall be sinusoidal and the frequency not over 100 cycles per second.

3.6 Service conditions. The equipment shall meet the following service conditions:

3.6.1 *Equipment nonoperating.* Except as stated in 3.6.1(b) below the equipment shall comply with the operational requirements of 3.6.2 after subjection to any of the following nonoperating conditions succesively or in combinations encountered during worldwide short-term storage and transit:

(a) *Temperature.* Exposure in the range of ±160°F. to —80°F.; however, exposure at the high temperature extreme not to exceed 4 hours and at the low temperature extreme not to exceed 24 hours at any one time.

(b) *Relative humidity.* Relative humidity up to 97 percent, for an indefinite period of time; and exposure at 100 percent relative humidity, with condensation, for 4 hours. Following subjection to such conditions and a power-on period of 16 hours at normal room temperature and relative humidity, the equipment shall meet the requirements of 3.6.2.

(c) *Orientation.* Indefinite storage in any orientation.

(d) *Bounce.* As specified in 3.7.

3.6.2 *Equipment operating.* After a warm-up period of 4 hours, the equipment shall meet the performance requirements of this specification while subjected to any of the following conditions successively or in combinations likely to be encountered during worldwide operations:

(a) *Temperature.* Ambient temperature in the range of + 150°F. to — 40°F. (The 150°F. temperature includes effect of sun-load.) Exposure at the high temperature extreme not to exceed 4 hours and at the low temperature extreme not to exceed 72 hours.

(b) *Relative humidity.* Relative humidity up to 97 percent and exposure at 100 percent relative humidity, with condensation, for 4 hours.

(c) *Elevation.* Elevation up to 10,000 feet above sea level.

(d) *Orientation.* Any orientation up to 20 degrees from normal operating orientation.

3.7 Bounce. The equipment shall meet full specification requirements after being subjected to the test specified in 4.6.3.

3.8 Preconditioning. The equipment shall

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be capable of meeting the inspection of section 4, without subsequent processing, after subjection to the bounce preconditioning of 4.4.

3.9 Radio interference suppression. The equipment shall conform to the requirements of Specification MIL–S–11748 and the interference limits set forth therein, with the antenna of the interference measuring equipment located 2 feet from the equipment.

3.10 Interchangeability. Corresponding units and replaceable assemblies and subassemblies and parts on contract shall be physically and functionally interchangeable as complete items without modification thereof or of other articles with which the items are used. When dimensions, ratings, characteristics, etc., are not specified, the manufacturer's design limits shall be used to determine compliance with the foregoing. If the contractor is in doubt as to whether a particular assembly, subassembly or part is to be considered replaceable, the contracting officer shall be consulted.

3.11 Identification of product. Equipment, assemblies and parts shall be marked for identification in accordance with Standard MIL–STD–130.

3.12 Workmanship. The equipment shall be manufactured and assembled in accordance with the applicable portions of the following paragraphs herein:

3.4 — Parts, materials, and processes.

3.4.2.2 — Riveting.

3.4.2.3 — Soldering.

3.4.2.4 — Welding.

3.4.2.5 — Wiring and cabling.

4. QUALITY ASSURANCE PROVISIONS

4.1 Classification of inspection. Inspection shall be classified as follows:

(a) Preproduction inspection (does not include preparation for delivery) (see 4.2).

(b) Procurement inspection. (Procurement inspection shall be the inspection performed by the contractor and by the Government, as specified in 4.3 and 4.12).

(1) Procurement inspection of items before preparation for delivery (see 4.3).

(2) Procurement inspection of preparation for delivery (see 4.12 and sec. 5).

4.2 Preproduction inspection. This inspection shall be performed by the Government unless otherwise specified in the contract. It shall consist of the preproduction inspection specified in table I, and the nondestructive examinations and tests listed in tables II and III. Sampling shall not be applicable in preproduction inspection. Other nondestructive inspection may be performed on preproduction samples to determine compliance with specified requirements.

TABLE I. *Preproduction inspection*

Inspection	Requirement par.	Inspection par.
Moisture resistance	3.6	4.6.1
Temperature	3.6	4.6.2
Bounce	3.7	4.6.3
Operational	...	4.7
Radio interference	3.9	4.8
Interchangeability	3.10	4.9
Welding	3.4.2.4	4.10
Workmanship	3.12	4.11

4.3 Procurement inspection of items before preparation for delivery. The minimum inspection performed by the contractor shall be as specified by (a) through (d) below — to demonstrate compliance with specified requirements. The Government will review and evaluate the contractor's inspection proce-

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dures and examine the contractor's inspection records. In addition, the Government — at its discretion — may verify, by product inspection, the contractor's compliance with specified requirements. Such verification shall not exceed the amount of inspection required herein of the contractor.

(a) Procurement inspection shall consist of the group A and group B inspection specified in 4.3.1 and 4.3.2.

(b) When Standard MIL–STD–105 specifies actions by the Government, the Government may authorize the contractor to perform any of such actions except that responsibility for acceptance rests with the Government.

(c) Group B inspection shall normally be performed on inspection lots that have passed group A inspection and on samples selected from units that have been subjected to and met the group A inspection.

(d) Each unit which will be subjected to the group A and group B inspection shall be preconditioned after final assembly (see 3.8).

4.3.1 *Group A inspection.* This inspection (including sampling) shall conform to table II and Standard MIL–STD–105. Unless otherwise specified, normal inspection shall be used at the start of the contract. Group A inspection shall be performed in any order which is satisfactory to the Government, except that the operational inspection shall be last.

TABLE II. *Group A inspection*

Inspection or test	Requirement paragraph	Inspection paragraph	AQL (dphu)	
			Major	Minor
Electrical:	3.5	4.5		*
A.c. ringing converter	3.5.1	4.5.1	6.5	
D.c. ringing converter	3.5.2	4.5.2	for the	
Circuit breakers	3.5.3	4.5.3	group	
Operational test		4.7		
Visual and mechanical	312	4.11	6.5	25

* All electrical defects are of major category.

4.3.2 *Group B inspection.* This inspection (including sampling) shall conform to table III and the appendix to Standard MIL–STD–105. Unless otherwise specified herein, normal inspection shall be used at the start of the contract. The reduced inspection procedure shall be R–1. Disposition of nonconforming product (sample units and inspection lots) shall be in accordance with 4.3.3 and the requirements of Standard MIL–STD–105 for disposition of rejected product.

4.3.2.1 *Group B sampling plans.* The AQL shall be 6.5 percent defective and the inspection level shall be I–7 for normal and tightened inspection and I–5 for reduced inspection.

4.3.2.2 *Order of inspection within group B.* Group B inspection shall be performed in any order which is satisfactory to the Government.

4.3.3 *Disposition of nonconforming product.* When defective sample units or failed inspection lots are submitted for acceptance, such product shall be suitably tagged or identified by equivalent means to indicate the cause of failure and means employed to correct the fault. The required information shall be presented to the Government.

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TABLE III. *Group B inspection*

Inspection	Requirement paragraph	Inspection paragraph	Sampling Plan
Breakdown voltage	3.5.4	4.5.4	6.5% Level 7 N and T Level 5 R

4.4 Bounce preconditioning. The equipment shall be placed in its normal operating position on the table of the package tester as made by L.A.B. Corporation, Skaneateles, New York, or equal. The package tester, shafts in phase, shall have a speed such that it is possible to insert a 1/32-inch-thick strip of material under one corner or edge of the equipment to a distance of 3 inches as the equipment bounces. The equipment shall be subjected to this preconditioning for 1 minute. After bounce preconditioning the equipment shall not be repaired, alined, cleaned or otherwise changed prior to subjection to acceptance inspection.

4.5 Electrical tests.

4.5.1 *ac ringing converter.* The equipment shall be tested to determine compliance with 3.5.1.

4.5.2 *dc ringing converter.* The equipment shall be tested to determine compliance with 3.5.2.

4.5.3 *Circuit breakers.* Circuit breakers shall be tested at 95° ± 5°F. as follows:

(a) Pass 100 percent rated current through the circuit breaker and a noninductive resistance load for not less than 10 seconds. The circuit breaker shall not "trip" or open its closed contacts.

(b) Pass 125 percent rated current through the circuit breaker and a noninductive resistance load. The circuit breaker shall "trip" to open closed contacts in 8 seconds (or less) for ac breakers and in 2 seconds (or less) for dc breakers.

(c) Pass 300 percent rated current through the circuit breaker and a noninductive resistance load. The circuit breaker shall "trip" to open closed contacts in 0.4 seconds (or less) for ac breakers and in 0.3 seconds (or less) for dc breakers.

4.5.4 *Breakdown voltage.* The equipment shall be tested to determine compliance with 3.5.4.

4.6 Service condition tests.

4.6.1 *Moisture-resistance.*

4.6.1.1 *Test conditions.*

(a) Do not move equipment from the humidity chamber for measurements.

(b) Start measurements not more than 5 minutes after power is applied to the equipment. Complete measurements as rapidly as possible. Do not leave power on after measurements have been completed.

4.6.1.2 *Test method.*

(a) Dry at 130° ± 5°F. for 24 hours.

(b) Condition at 77° ± 5°F. and 40 to 50 percent relative humidity for 24 hours.

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(c) Take measurements as specified in 4.6.1.3 and readjust or realine as necessary to meet full specification requirements.

(d) Subject to continuous cycling for five 48-hour cycles. Temperature, relative humidity and period of time for each portion of the cycle shall conform to Standard MIL–STD–170. The measurements specified in 4.6.1.3 shall be made during the periods shown on the standard and shall comply with the performance during cycling specified in 4.6.1.3.

(e) After cycling has been completed, condition the equipment for 24 hours at 77° ± 5°F. and 40 to 60 percent relative humidity. Then adjust for optimum performance using only those means provided by the equipment. No repair or replacement of parts shall be made. After adjustment, the equipment shall meet full specified performance for those measurements specified in 4.6.1.3.

4.6.1.3 *Performance.* The equipment shall be capable of meeting the tests of 4.5 and 4.11 after completion of the moisture resistance test.

4.6.1.4 *Failure.* If any equipment fails to meet the performance specified in 4.6.1.3 during cycling, it shall be realined or readjusted once. If the equipment then fails to meet 4.6.1.3 or fails subsequently during cycling, it does not pass the test. In addition, if the equipment fails to meet full specification requirements after conditioning and adjustment, as specified in 4.6.1.2(e), it does not pass the test.

4.6.2 *Temperature.* The equipment shall be subjected to the temperature cycle shown on Standard MIL–STD–169. The operational test specified in 4.7 shall be made at step 3 and step 9 indicated on the standard.

4.6.3 *Bounce.* The equipment shall be placed on the table of the package tester as made by L.A.B. Corporation, Skaneateles, New York, or equal, and shall be constrained from horizontal motion of more than 2 inches by suitable wooden fences. The package tester, shafts in phase, shall be operated at a speed of 285 rpm ± 1.0 percent, for a total of 3 hours. During each ½ hour period of the test, the transit case (with contained equipment) shall rest on a different face.

4.7 Operational test. The equipment shall be tested to insure the proper operation of all component parts and circuits as follows:

4.7.1 *Preparation.* Connect 110 volt, 60 cycle, ac power to "IN–110 V" receptacle and a 48-volt storage battery to the IN–B receptacle.

4.7.2 *Power circuits.*

(a) *Preparation.*

(1) 110 V circuit breaker switch "ON".

(2) 48 V circuit breaker switch "ON".

(3) ac switch of rectifier "OFF".

(4) dc switch of rectifier "OFF".

(5) Fine adjustment control and coarse adjustment control switches on step 1.

(b) *Test procedure.*

(1) Place ac switch of rectifier to "ON". The red lamp should light and rectifier meter read zero.

(2) Place dc switch of rectifier to "ON". Rectifier and panel ammeters shall indicate current flow changes. Opera-

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tion of adjustment control switches shall show increases of current in both ammeters.

(3) With a portable dc voltmeter, test voltages between the plus bar or board ground terminal and minus terminals of the 48 volt binding post and receptacle; also, test voltages between the minus bar and plus terminals of 48 volt binding post and receptacle. The readings at the binding post shall be in accordance with the voltages across the bus bar.

(4) A 110 volt extension lamp shall operate when tested in the 110 volt outlet receptacles.

(5) Connect a test lead between BAT binding posts. Test with voltmeter between minus bus bar and VT terminal of receptable number 203. The voltmeter shall register battery voltage.

4.7.3 *Ringing circuits.*

(a) *Test procedure.*

(1) 110 volt circuit breaker switch "ON".

(2) Telering emergency ring switch on "TELERING".

(3) Telering switch "ON".

(4) With Telephone EE–8, or equal, connected to the 20 cycle outlet, telephone shall ring.

(b) *Test procedure.*

(1) 48-volt circuit breaker "ON".

(2) Connect voltmeter test leads between minus bus bar and long prong of 20-cycle outlet. The voltmeter shall read the voltage of the battery.

(3) Telering-emergency ring switch on emergency ring and emergency ring circuit breaker switch "ON".

(4) With Telephone EE–8, or equal, connected to either one of the 20-cycle outlets, telephone shall ring.

4.8 Radio interference suppresion test. The equipment shall be tested for compliance with 3.9.

4.9 Inspection for interchangeability. Replaceable parts, subassemblies, and components, shall be inspected to determine compliance with the physical interchangeability requirements of 3.10. Replaceable parts, subassemblies, and components, which do not meet the requirements of 3.10 shall be considered a major defect.

4.10 Welding. Welding shall be inspected by the magnetic particle method specified in Specification MIL–M–11473 and by the ultrasonic method using a Reflectoscope, as made by Sperry Products, Incorporated, Danbury, Connecticut, or equal. Both tests shall be performed on each sample unit.

4.11 Visual and mechanical inspection. The equipment shall be examined for the defects listed in Standard MIL–STD–252 and to determine compliance with the requirements of 3.12.

4.12 Procurement inspection of preparation for delivery. The minimum inspection performed by the contractor shall be as specified in 4.12.1 and 4.12.2. The Government will review and evaluate the contractor's inspection procedures and examine the contractor's inspection records. In addition, the Government — at its discretion — may

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verify, by product inspection, the contractor's compliance with specified requirements. Such verification shall not exceed the amount of inspection required herein of the contractor. When Standard MIL–STD–105 specifies action by the Government, the Government may authorize the contractor to perform any of such actions except that responsibility for acceptance rests with the Government.

4.12.1 *Preservation and packaging.* Inspection of preservation and packaging shall be as specified in Specification MIL–P–116. Classification of defects shall conform to tables IV and V.

4.12.2 *Packing and marking of exterior containers.* Packing and marking of exterior containers shall be given visual inspection for the defects listed in tables IV and V and to determine conformance with the approved process sheet furnished by the contractor as required by the contract. This inspection shall conform to the appendix to Standard MIL–STD–105. Inspection level L–8 shall be used for normal inspection and L–6 for reduced inspection. Unless otherwise specified herein, normal inspection shall be used at the start of the contract. The reduced inspection procedure shall be R–1. The AQL for major defects shall be four percent and the AQL for minor defects shall be ten percent.

4.12.3 *Inspection lot.* A lot for visual inspection of the pack shall be all completed packs which are identical and are submitted for inspection at one time.

4.12.4 *Procedure in case of failure.* If an inspection lot is rejected the contractor shall immediately investigate the cause of failure and shall report to the Government inspector the results thereof and details of the corrective action taken. If the contractor and the Government inspector cannot agree on the effectiveness of the corrective action, the matter shall be referred to the contracting officer for resolution.

4.12.5 *Disposition of nonconforming product.* Disposition of rejected lots and defective sample units shall be in accordance with Standard MIL–STD–105. When resubmitted for acceptance, such product shall be suitably tagged or identified by equivalent means to indicate the cause of failure and means employed to correct the fault. The required information shall be presented to the Government when the product is resubmitted.

TABLE IV. *Classification of defects for preservation, packaging and marking*

Major defects

1. Use of improper or defective material.
2. Quantity in unit package not as specified.
3. Incorrect packaging method applied.
4. Cushioning or padding omitted.
5. Cushioning inadequate for the physical and mechanical protection of the item.
6. Unsealed, punctured, or improperly sealed barrier bag, wrap, or envelope.
7. Stock number omitted, incorrect or illegible.
8. Nomenclature omitted, incorrect, or illegible.
9. Marking of quantity of items in package omitted, incorrect, or illegible.
10. Different stock numbered items in the same unit package.

Minor defects

1. Item not properly blocked or braced within the unit package to prevent movement.
2. Packaging material damaged.
3. Any item of marking information other than 7 thru 9 omitted, incorrect, or illegible.

TABLE V. *Classification of defects for packing and marking*

Major defects

1. Use of improper or defective material.
2. Box closure not as specified.
3. Type, grade, class, and style of the shipping container not as specified.

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4. Strapping omitted.
5. Strapping inadequate or incorrectly applied.
6. Items not adequately blocked, braced, or cushioned within the shipping container to prevent movement or damage.
7. Shipping documents or packing list omitted.
8. Stock number omitted, incorrect, or illegible.
9. Nomenclature omitted, incorrect, or illegible.
10. Marking of quantity of items in pack omitted, incorrect, or illegible.
11. Destination marking omitted, incorrect, or illegible.
12. Service designation (color marking) omitted.
13. Specified special marking and labeling such as MAP labels, shipment digit markings, etc., omitted, incorrect, or illegible.
14. Overseas code marking omitted, incorrect, or illegible.

Minor defects

1. Unsealed carton.
2. Defective taping or sealing of carton.
3. Any other box defect which may be considered minor by definition of Standard MIL-STD-105.
4. Any item of required marking information other than 8 thru 14 listed under major defects omitted, incorrect, or illegible.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.

5.1.1 *Level A.* Panel, Telephone Switchboard (Panel BD–132( )) shall be preserved, packaged an tested in accordance with the procedures specified for the designated methods as prescribed in Specification MIL–P–116 as described herein.

5.1.1.1 *Technical manual.* Each technical manual shall be packaged Method IC–3 as follows: Inclose each technical manual within a close-fitting type I, class B bag conforming to Specification MIL–B–117. Heat seal to effect closure.

5.1.1.2 *Portfolio of circuit description and drawings.* No packaging required.

5.1.1.3 *Lamps and fuses (spares).* Spare lamps and fuses shall be packaged Method III as follows: Items of like description shall be grouped together and each group shall be placed within a wrap of single faced flexible corrugated fiberboard, type I, conforming to Specification LLL–F–291. Secure the wrap with class 1 gummed paper tape conforming to Specification UU–T–111.

5.1.1.4 *Panel, telephone, switchboard (Panel BD–132( )).*

5.1.1.4.1 *Prepackaging.* All components within the front of the panel shall be blocked and braced to prevent movement and shearing stress on the mountings by means of wood blocking or cells or pads or both fabricated of double faced corrugated fiberboard having a minimum dry bursting strength of 275 pounds in conformance with the applicable provisions of table I, Specification PPP–B–636. Wood blocking of contacting critical surfaces of any component shall be cushioned with grade C, class 1 barrier material conforming to Specification MIL–B–121. Place the items packaged as specified in 5.1.1.2 and 5.1.1.3 within the spare parts box located in the front cover of the panel. Position the front and rear covers of the panel and secure all fastenings.

5.1.1.4.2 *Packaging.* Each Panel BD–132( ) shall be packaged Method IC–5 as follows: Cushion the four longest edges of the panel with double fold pads fabricated of double faced corrugated fiberboard having a minimum dry bursting strength of 200 pounds conforming to table I, Specification PPP–B–636. Fabricate a close fitting sleeve of corrugated fiberboard, type I, class 2, grade 3 (V3c), table II, Specification PPP–B–636, and two end caps (one for each end) of half slotted containers, to completely inclose the cushioned panel (see figure 1). Closure shall be in conformance with the require-

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ments of the appendix of the box specification. In addition, seal all seams and joints with waterproof, pressure-sensitive tape, 3 inches wide, type II, class 1, olive drab, conforming to Specification PPP–T–60.

5.1.2 Level C. Panel BD–132( ) shall be packaged in accordance with commercial practice and in a manner that will afford protection against corrosion, deterioration and physical damage during direct shipment to the first receiving activity.

5.2 **Packing.**

5.2.1 *Level A.* Panel BD–132( ) shall be packed for shipment within a style 2, nailed wood box conforming to the requirements of Specification PPP–B–621, modified to include a skid type base, and interior diagonal bracing (see figure 2). The size of the cleats and interior bracing members shall be determined by limitations of weight and volume. The sheathing parallel to the skids shall run vertically to permit nailing into the skids. Fabricate the box to fit the contents snugly. Place the technical manual (see 5.1.1.1) between the packaged item and lid of the shipping container. Box closure shall be in conformance with the box specification .

5.2.1.1 *Metal strapping.* Shipping containers shall be strapped with flat metal strapping in conformance with the requirements of the appendix of the container specification, only for direct shipment to ports.

5.2.2 *Level B.* Panel BD–132( ) shall be packed for shipment as specified in 5.2.1.

5.2.3 *Level C.* Panel BD–132( ) shall be packed for shipment in a manner conforming to the requirements of Uniform Freight Classification for rail shipment, National Motor Freight Classification for truck shipment, and the regulations of other carriers as applicable to the mode of transportation employed at the lowest transportation rate.

5.3 **Marking.** Interior packages and exterior shipping containers shall be marked in accordance with the applicable provisions of Standard MIL–STD–129.

6. NOTES

6.1 **Intended use.** Panel BD–132( ) is the terminal point for protection, control and distribution of power and ringing leads which supply a telephone central office.

6.2 **Ordering data.**

6.2.1 *General requirements.*

(a) Title, number, and date of this specification.

(b) Whether domestic or overseas shipment is required.

(c) Quantity of tools and running spare parts required.

(d) Technical literature required.

(e) Facsimiles required.

(f) Serial numbers.

(g) Preproduction samples required.

6.2.2 *Preparation for delivery.*

(a) Level of packaging and level of packing required for shipment (level A, level B, or level C).

(b) Preproduction pack(s) as follows:

(1) Makeup of pack(s).

(2) Number of each kind of pack to be submitted.

(3) Inspection that will be performed thereon — including rough handling test, which will not be performed as acceptance inspection.

(4) Five copies of process sheet, using format shown on figure 3, and accompanied by such drawings, sketches, and other data as may be neces-

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sary to completely describe the procedure followed in fabrication of the preproduction pack(s) and to identify the items therein. These data should be submitted to the contracting officer with the preproduction pack.

(5) A packaging technician from the procuring agency will evaluate the preproduction pack(s).

(c) The specific paragraphs of section 5 which are applicable to the particular procurement.

6.3 Nomenclature. The parentheses in the nomenclature will be deleted or replaced by a letter identifying the particular design; for example: BD–132–W. As soon as possible after the award of the contract, the contractor should apply to the Government office specified in the contract for such information.

6.4 Definitions.

6.4.1 *Inspection.* Inspection is the examination or testing, or both, of supplies to determine compliance with applicable requirements. Sampling is an element of inspection.

6.4.2 *Examinations.* Examination consists of simple, generally nondestructive determinations of compliance, without use of special testing equipment.

6.4.3 *Testing.* Testing consists of determinations of compliance, using technical means.

Notice. When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

Custodians:
Army—Signal Corps
Navy—Bureau of Yards and Docks
Air Force

Other interest:
Navy—MC

Preparing activity:
Army—Signal Corps

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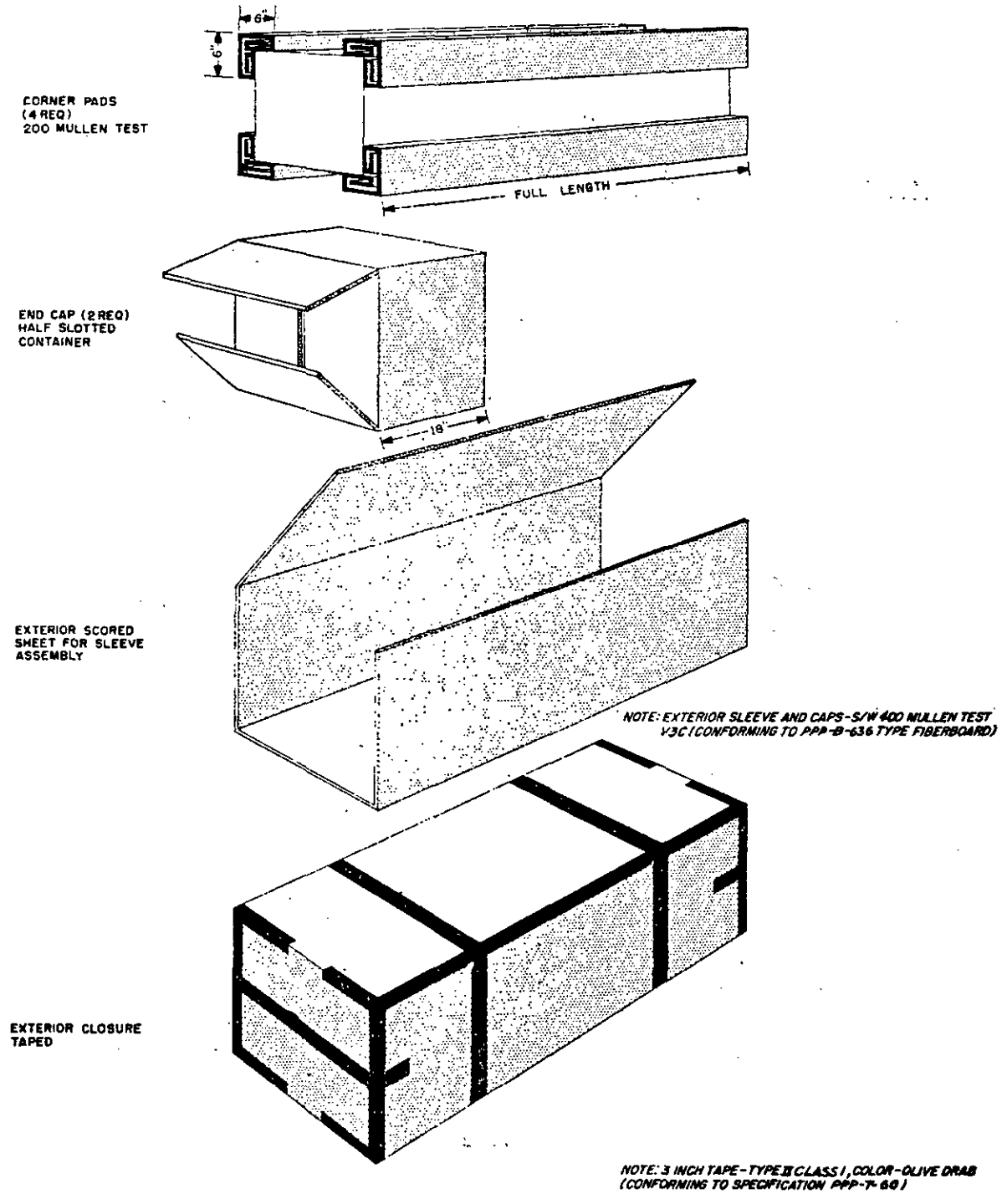
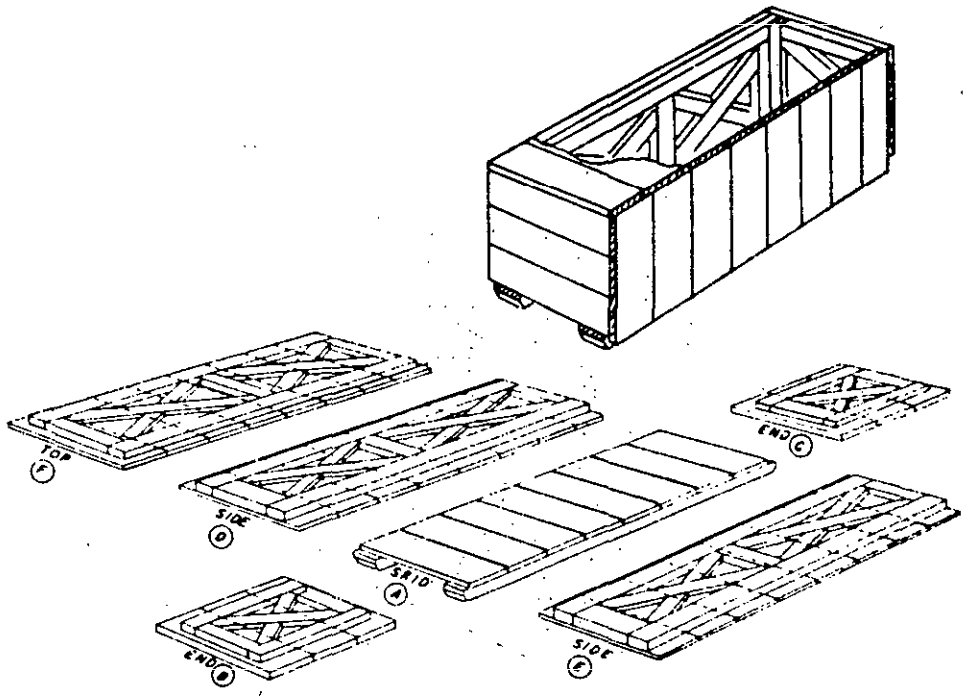


FIGURE 1

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MODIFIED STYLE 2

"XX" PANEL CONSTRUCTION:



1. The skid Ⓐ sides Ⓓ Ⓔ ends Ⓑ Ⓒ and top Ⓕ shall be prefabricated as independent units as illustrated.

2. Assemble prefabricated units as follows.

NAIL Ⓓ & Ⓔ TO Ⓐ
NAIL Ⓑ & Ⓒ TO ⒶⒹ & Ⓔ
NAIL Ⓕ TO ⒷⒸⒹⒺ

FIGURE 2

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8½" OR LARGER, AS REQUIRED

PACKAGING, PACKING, AND MARKING PILOT PACK PROCESS SHEET

CONTRACTOR
STREET ADDRESS
CITY AND STATE

ORDER NO. _____
EQUIPMENT _____
LEVEL _____

DATE PILOT PACK COMPLETED FOR EVALUATION

ITEM NAME								
PACKAGING METHOD								
CLEANING	SPECIFICATION							
	MATERIAL DESCRIPTION	T	G	C	T		G	
PRESERVATION	SPECIFICATION							
	MATERIAL DESCRIPTION	T	G	C	T		G	
CUSHIONING	SPECIFICATION							
	MATERIAL DESCRIPTION	T	G	C	T		G	
	SIZE							
SECURING	SPECIFICATION							
	MATERIAL DESCRIPTION	T	G	C	T		G	
	SIZE							
BARRIER	SPECIFICATION							
	MATERIAL DESCRIPTION	T	G	C	T		G	
	SIZE							
DESICCANT	SPECIFICATION							
	MATERIAL DESCRIPTION	T	G	C	T		G	
	SIZE							
UNIT CARTON	SPECIFICATION							
	MATERIAL DESCRIPTION	T	G	C	T		G	
	SIZE							
INTERMEDIATE CARTON	SPECIFICATION							
	MATERIAL DESCRIPTION	T	G	C	T		G	
	SIZE							
OUTER CARTON (CONSOLIDATED OR IIb)	SPECIFICATION							
	MATERIAL DESCRIPTION	T	G	C	T		G	
	SIZE							
SECURE CLOSURE	SPECIFICATION							
	MATERIAL DESCRIPTION	T	G	C	T		G	
	SIZE							

USE AS MANY COLUMNS AS NECESSARY

11"

CASE LINER	SPECIFICATION	
	SEALING MATERIAL	
	SIZE	
SHIPPING CONTAINER	SPECIFICATION	
	STYLE	
	SIZE	
STRAPPING	SPECIFICATION	
	MATERIAL DESCRIPTION	
	SIZE	

LEGEND FOR MATERIAL: T - TYPE, G - GRADE, C - CLASS

PAGE OF PAGES

FIGURE 8

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